2. In general to find the common difference of an A.P. we should subtract first term from the second term, second from the third and so on.

For example, $t_1 = a,\ t_2 = a + d$

$\therefore\ t_2 - t_1 = (a + d) - a = d$

Similarly, $t_2 = a + d,\ t_3 = a + 2d, \ldots$

$\therefore\ t_3 - t_2 = (a + 2d) - (a + d) = d$

In general, $d = t_2 - t_1 = t_3 - t_2 = t_4 - t_3 = \ldots.$

$d = t_n - t_{n-1}$ for $n = 2, 3, 4, \ldots$

Progress Check

1. The difference between any two consecutive terms of an A.P. is _____.
2. If a and d are the first term and common difference of an A.P. then the 8th term is _____.
3. If t_n is the nth term of an A.P., then $t_{2n} - t_n$ is _____.

Let us try to find the common differences of the following A.P.'s

(i) $1, 4, 7, 10, \ldots$

$d = 4 - 1 = 7 - 4 = 10 - 7 = \ldots = 3$

(ii) $6, 2, -2, -6, \ldots$

$d = 2 - 6 = -2 - 2 = -6 - (-2) = \ldots = -4$

Do You Know?

The common difference of an A.P. can be positive, negative or zero.

Thinking Corner

If t_n is the nth term of an A.P. then the value of $t_{n+1} - t_{n-1}$ is _____.

Example 2.23 Check whether the following sequences are in A.P. or not?

(i) $x + 2,\ 2x + 3,\ 3x + 4, \ldots$ (ii) $2, 4, 8, 16, \ldots.$ (iii) $3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, 9\sqrt{2}, \ldots$

Solution To check that the given sequence is in A.P., it is enough to check if the differences between the consecutive terms are equal or not.

(i) $t_2 - t_1 = (2x + 3) - (x + 2) = x + 1$

$t_3 - t_2 = (3x + 4) - (2x + 3) = x + 1$

$t_2 - t_1 = t_3 - t_2$

Thus, the differences between consecutive terms are equal.

Hence the sequence $x + 2,\ 2x + 3,\ 3x + 4, \ldots$ is in A.P.

(ii) $t_2 - t_1 = 4 - 2 = 2$

$t_3 - t_2 = 8 - 4 = 4$

$t_2 - t_1 \neq t_3 - t_2$

Thus, the differences between consecutive terms are not equal. Hence the terms of the sequence $2, 4, 8, 16, \ldots$ are not in A.P.

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